

Code changes from implicit to CEL

fe-large-displacement

Each script in `examples/` is complete. Open two and diff. This note lists the extra lines at each rung.

1 Implicit

`examples/implicit.py`

Geometrically nonlinear static step

Automatic stabilisation is fictitious viscous damping. Check ALLSD.

```

120 # *Static, geometrically nonlinear, automatic stabilisation (fictitious
121 # viscous damping). Check ALLSD against ALLIE if the job runs far enough.
122 model.StaticStep(
123     name='Push', previous='Initial', nlgeom=ON, timePeriod=1.0,
124     initialInc=0.005, minInc=1.0e-8, maxInc=0.05, maxNumInc=1000,
125     stabilizationMethod=DISSIPATED_ENERGY_FRACTION,
126     stabilizationMagnitude=2.0e-4, adaptiveDampingRatio=0.05,
127     continueDampingFactors=ON)
128 model.TabularAmplitude(name='Ramp', timeSpan=STEP,
129     data=((0.0, 0.0), (1.0, 1.0)))

```

Displacement control

```

157
158 # Displacement control: push the rigid body down by IND_DEPTH over the step.
159 model.DisplacementBC(
160     name='Push', createStepName='Push', region=asm.sets['IndRP'],
161     u1=0.0, u2=0.0, u3=-IND_DEPTH, ur1=0.0, ur2=0.0, ur3=0.0,

```

Standard element library

```

96 # First-order reduced-integration hex, Standard library.
97 elem = mesh.ElemType(elemCode=C3D8R, elemLibrary=STANDARD)

```

Standard general contact

```

168     formulation=PENALTY, directionality=ISOTROPIC, table=((FRICTION,)),
169     maximumElasticSlip=FRACTION, fraction=0.005)
170 contact = model.ContactStd(name='GeneralContact', createStepName='Push')
171 contact.includedPairs.setValuesInStep(stepName='Push', useAllstar=True)

```

2 Implicit ALE

On top of implicit. `examples/implicit_ale.py`

Restricted adaptive domain

ALE only accepts first-order reduced-integration solids (C3D8R). An empty set is silent. Standard has no initial mesh sweeps.

```

166 # --- ALE (the only extra block versus implicit.py) -----
167 # Restrict the domain to a box around the indenter. An empty set is silent:
168 # Abaqus will not warn, and ALE will do nothing.
169
170 half = 3.0 * IND_HALF
171 zmin = max(-1.3 * IND_DEPTH, -SOIL_DEP + 2.0 * SEED)
172 ale_elems = soil_inst.elements.getByBoundingBox(
173     xMin=-half, xMax=half, yMin=-SOIL_WID, yMax=SOIL_WID,
174     zMin=zmin, zMax=0.0)
175 if len(ale_elems) == 0:
176     raise RuntimeError('ALE box is empty. ALE would run and do nothing.')
177 asm.Set(name='ALE_Box', elements=ale_elems)
178 print('ALE elements: %d' % len(asm.sets['ALE_Box'].elements))
179
180 model.AdaptiveMeshControl(name='ALE_BC', smoothingPriority=UNIFORM)
181 # Standard has no initial mesh sweeps. Do not pass initialMeshSweeps.
182 model.steps['Push'].AdaptiveMeshDomain(
183     region=asm.sets['ALE_Box'], controls='ALE_BC',
184     frequency=10, meshSweeps=1)

```

Diagnostics after *Static

CAE cannot write *Diagnostics. Without it you cannot prove ALE ran.

```

196 # CAE cannot write *Diagnostics. Insert it after *Static.
197 _inject_diagnostics(inp, after_static=True)

```

3 Explicit

On top of implicit, replacing the solver. `examples/explicit.py`

Mass (*Density)

Required. The stable increment is $L_{\min}/c_d$.

```

79 soil_mat.Density(table=((SOIL_RHO,)),)
80 soil_mat.Elastic(table=((SOIL_E, SOIL_NU),))
81 steel = model.Material(name='SteelMat')

```

Explicit element library

```

94 # Same C3D8R, Explicit library.
95 elem = mesh.ElemType(elemCode=C3D8R, elemLibrary=EXPLICIT)

```

Dynamic explicit step, not static

Drop automatic stabilisation. Bulk viscosity is on by default (ALLVD). Do not add extra damping to hide kinetic energy.

```

118 # *Dynamic, Explicit. No automatic stabilisation. Bulk viscosity is on by
119 # default (watch ALLVD). Do not add extra damping to hide kinetic energy.
120 time_period = IND_DEPTH / float(IND_VEL)
121 model.ExplicitDynamicsStep(
122     name='Push', previous='Initial', timePeriod=time_period,
123     nlgeom=ON, improvedDtMethod=ON)
124 model.SmoothStepAmplitude(
125     name='Ramp', timeSpan=STEP,

```

```
126 data=((0.0, 0.0), (RAMP_FRAC * time_period, 1.0))
```

Energy history

Peak ALLKE/ALLIE below about 5 % is acceptable, below 1 % is quotable.

```
127 model.HistoryOutputRequest(
128     name='Energies', createStepName='Push',
129     variables=('ALLIE', 'ALLKE', 'ALLAE', 'ALLDC', 'ALLVD', 'ALLWK',
130               'ETOTAL'),
131     numIntervals=200)
```

Velocity control, not displacement

```
156 # Velocity control, not displacement. Sign is negative: penetration is -z.
157 model.VelocityBC(
158     name='Push', createStepName='Push', region=asm.sets['IndRP'],
159     v1=0.0, v2=0.0, v3=-IND_VEL, vr1=0.0, vr2=0.0, vr3=0.0,
160     amplitude='Ramp', distributionType=UNIFORM)
```

Explicit general contact

```
168 contact = model.ContactExp(name='GeneralContact', createStepName='Push')
169 contact.includedPairs.setValuesInStep(stepName='Push', useAllstar=True)
170 contact.contactPropertyAssignments.appendInStep(
171     stepName='Push', assignments=((GLOBAL, SELF, 'Interface')))
```

Double precision job

```
177 #      1, 0, 0.0, 0.0, 0, None),))
178
179
180 # --- write, check, maybe submit -----
181
182 job = mdb.Job(
```

Optional mass scaling

Commented in examples/explicit.py. Semi-automatic, target Δt . Then re-check ALLKE.

4 Explicit ALE

On top of explicit. examples/explicit_ale.py

Adaptive domain with initial sweeps

Same box as implicit ALE. Explicit adds initialMeshSweeps=5.

```
166
167
168 # --- ALE (the only extra block versus explicit.py) -----
169
170 half = 3.0 * IND_HALF
171 zmin = max(-1.3 * IND_DEPTH, -SOIL_DEP + 2.0 * SEED)
172 ale_elems = soil_inst.elements.getByBoundingBox(
173     xMin=-half, xMax=half, yMin=-SOIL_WID, yMax=SOIL_WID,
174     zMin=zmin, zMax=0.0)
175 if len(ale_elems) == 0:
176     raise RuntimeError('ALE box is empty. ALE would run and do nothing.')
```

```

177 asm.Set(name='ALE_Box', elements=ale_elems)
178 print('ALE elements: %d' % len(asm.sets['ALE_Box'].elements))
179
180 model.AdaptiveMeshControl(name='ALE_BC', smoothingPriority=UNIFORM)
181 # Explicit has initial mesh sweeps. Default frequency=10, meshSweeps=1.
182 model.steps['Push'].AdaptiveMeshDomain(
183     region=asm.sets['ALE_Box'], controls='ALE_BC',
184     frequency=10, meshSweeps=1, initialMeshSweeps=5)

```

Diagnostics after *Dynamic, Explicit

Then read the .msg. Zero percent of nodes moved means the domain was defined and did nothing.

```

198 # CAE cannot write *Diagnostics. Insert it after *Dynamic, Explicit.
199 _inject_diagnostics(inp)

```

Run one CPU for a study of ALE. Nodes shared between parallel domains are nonadaptive.

5 Explicit CEL

On top of explicit, replacing the Lagrangian soil. `examples/explicit_cel.py`

Eulerian part with void above the soil

Do not set the void to zero. Heave needs empty elements.

```

63 # Eulerian box: soil depth PLUS empty space above it for heave.
64 H = VOID
65 sketch = model.ConstrainedSketch(name='eul', sheetSize=10.0 * SOIL_LEN)
66 sketch.rectangle(point1=(-SOIL_LEN / 2.0, -SOIL_WID / 2.0),
67                 point2=(SOIL_LEN / 2.0, SOIL_WID / 2.0))
68 eul = model.Part(name='Eulerian', dimensionality=THREE_D, type=EULERIAN)
69 eul.BaseSolidExtrude(sketch=sketch, depth=SOIL_DEP + H)
70
71 # Partition at the soil surface so the filled region is a cell, not a guess.
72 if H > 0.0:
73     datum = eul.DatumPlaneByPrincipalPlane(principalPlane=XYPLANE,
74                                             offset=SOIL_DEP)
75     eul.PartitionCellByDatumPlane(datumPlane=eul.datums[datum.id],
76                                 cells=eul.cells)
76

```

Eulerian section and EC3D8R

```

92 # Eulerian section, not a solid section. The instance name is the EVF handle.
93 eul_key = 'soilmat-1'
94 model.EulerianSection(name='EulSec', data={eul_key: 'SoilMat'})
95 eul.SectionAssignment(region=regionToolset.Region(cells=eul.cells),
96                     sectionName='EulSec')
97 model.HomogeneousSolidSection(name='IndSec', material='SteelMat')
98 indenter.SectionAssignment(region=regionToolset.Region(cells=indenter.cells),
99                          sectionName='IndSec', offsetType=MIDDLE_SURFACE,
100                          thicknessAssignment=FROM_SECTION)
101
102 eul.setElementType(
103     regions=(eul.cells,),
104     elemTypes=(mesh.ElemType(elemCode=EC3D8R, elemLibrary=EXPLICIT),))

```

Initial volume fraction

Cells below $z = 0$ start full. This is *Initial Conditions, `type=VOLUME FRACTION`.

```

124 # Cells below z = 0 start full. Everything else starts empty.
125 filled = eul_inst.cells.getByBoundingBox(
126     zMin=-SOIL_DEP - 1e-6, zMax=0.0 + 1e-6)
127 if len(filled) == 0:
128     raise RuntimeError('no Eulerian cells below z = 0')
129 filled_set = asm.Set(name='SoilFilled', cells=filled)
130 model.MaterialAssignment(
131     name='Eulerian material assignment',
132     instanceList=(eul_inst,),
133     useFields=False,
134     assignmentList=((filled_set, (1.0,)),))

```

Walls, free top

Zero normal motion on the sides and base. The top is open.

```

166 # Walls: zero normal motion on sides and base. The top is open.
167 top = H
168 for name, point, dof in (
169     ('XMIN', (-SOIL_LEN / 2.0, 0.0, (top - SOIL_DEP) / 2.0), 1),

```

No geostatic, no ALE

*Geostatic has no Eulerian implementation. General contact is mandatory. There is no
 *Adaptive Mesh block.

6 What to check in the deck

Rung	The deck must show
Implicit	*Static, nlgeom=YES, displacement BC
Implicit ALE	plus *Adaptive Mesh on ALE_Box
Explicit	*Density, *Dynamic, Explicit, type=VELOCITY, ALLKE
Explicit ALE	plus *Adaptive Mesh, *Diagnostics
Explicit CEL	*Eulerian Section, EC3D8R, volume fraction, void, no ALE

Write the input. Read it. Then submit.